

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech, Summer 2025-26 Examination

Semester: VI/VII

Subject Code: 303105341

Subject Name: Cyber Security

Date: 17-04-2026

Time: 2:00 PM To 4:30 PM

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define stateful and stateless firewalls. b) Mention one key difference between stateful and stateless firewalls. c) Explain why stateful firewalls offer better security in tracking sessions.	06	CO3	BT2
	B. a) What is Netcat, and how is it used in network security? b) Mention two scanning functions of Netcat. c) Provide one example of a Netcat command used for vulnerability scanning.	06	CO2	BT4
Q.2	A. What is hairpin NAT regarding port forwarding?	04	CO3	BT1
	B. What is the default action taken by a firewall if no rule matches a packet?	05	CO3	BT1
	OR			
	B. Discuss the role of software testing in information system development.	05	CO1	BT4
Q.3	A. Illustrate how can captcha help mitigate automated dos attacks? with a relevant example.	04	CO3	BT3
	B. What steps can organizations take to stop data breaches?	05	CO3	BT1
	OR			
	B. Discuss the advantages and disadvantages of which type of malware self-replicates and spreads across networks without human intervention?	05	CO4	BT2

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define ransomware and its function. b) Mention two effects of ransomware on computers. c) Explain how ransomware attacks impact business operations.	06	CO4	BT2
	B. a) Define information security. b) Explain the principle of confidentiality in the CIA triad. c) Briefly describe the principles of integrity and availability in the CIA triad.	06	CO1	BT1
Q.2	A. What are the limitations of using a single tool for vulnerability scanning, and why is it important to use a combination of tools?	04	CO5	BT2

	B. What are the risks associated with leaving unnecessary ports open in a network, and how can vulnerability scanning mitigate these risks?	05	CO4	BT2
	OR			
	B. Explain the potential security risks associated with services running on open ports, and how vulnerability scanning can address these risks.	05	CO2	BT4
Q.3	A. What is the role of spyware in cybercrime? How can users stop infections?	04	CO4	BT6
	B. Clarify the idea of incident escalation when a security issue arises?	05	CO6	BT1
	OR			
	B. Discuss the advantages and disadvantages of information systems.	05	CO1	BT4